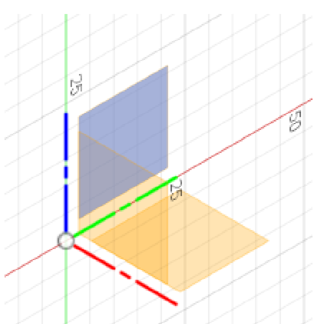
Creating and Documenting a Part in Fusion 360

Fusion 360 is a 3D modeling software that provides computer-aided design, computer-aided manufacturing, and computer-aided engineering capabilities all in one program. Fusion 360 is most frequently used by engineers to create sketches, visualize them in three dimensions, and modify them in a solid form. In addition to just creating parts, engineers use Fusion 360 to professionally document their parts so that they can sold, recreated, or mass produced. In this tutorial, you will be introduced to the fundamental processes of sketching a part, turning it into something three dimensional, modifying and detailing the part, and documenting it in a technical and professional way.

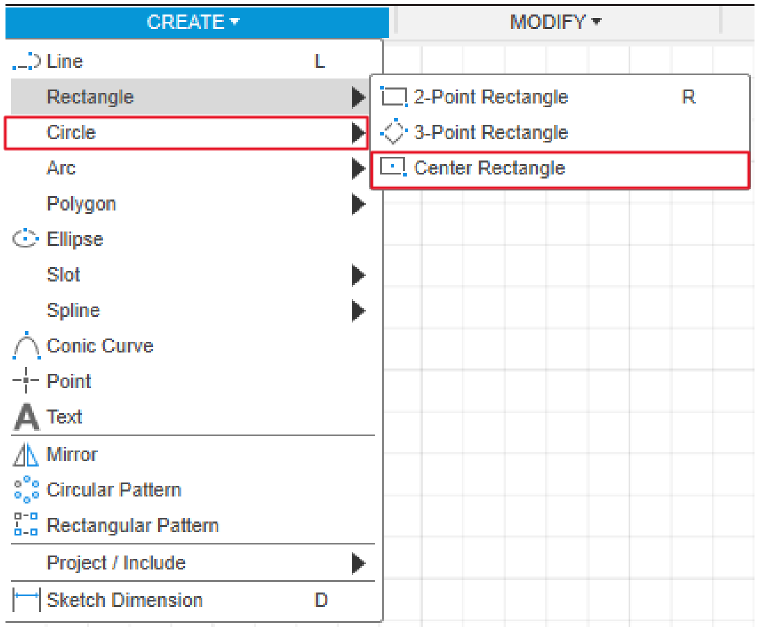
Creating a Sketch

The sketch is the foundation of your design. Using the sketch tool, you are able to create the geometric profile of what you will later turn into a three-dimensional object. The sketch tool gives you the ability to create rectangles, circle, polygons, ellipses, and even draw your sketch line by line. For this tutorial, we will be using the **Center Rectangle.**

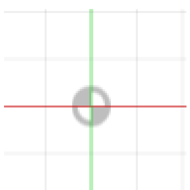
1. Open Fusion 360
2. On the left side of the home screen, click New. This will create a new file to begin your sketch.
3. Click **Create Sketch** under the **Solid** tab, then click a plane to begin the sketch.



4. At the top of the screen, click the Create dropdown → Rectangle → Center Rectangle.

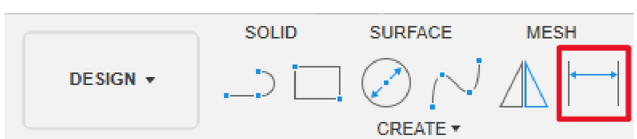


5. Click the Origin. This will place the center of the rectangle.

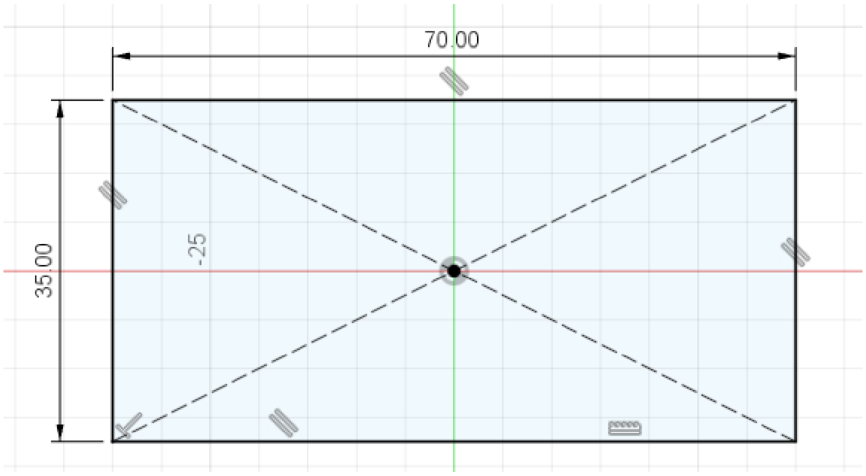


➢ *Note: Beginning your sketches from the origin can make future modifications to your sketches easier, although it is not necessary.*

6. With your cursor, drag the rectangle out an arbitrary amount and click – the exact size is not important for this step as we apply specific dimensions next.
7. Click Sketch Dimension or press D.

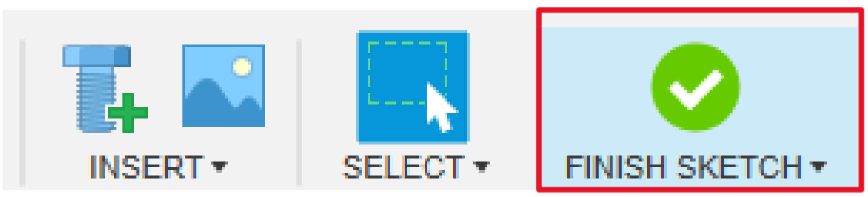


8. Click the top of the rectangle. Drag your cursor slightly up until you see a double-sided arrow between two vertical lines. Type '70' and hit Enter**.**
9. Click the left edge of the rectangle, type '35' and hit Enter. All the edges your center rectangle will be black.



Tip: When all edges of a sketch are black, that indicates it is fully constrained. A fully constrained sketch is considered good practice as it prevents unexpected movement of your sketches – this is especially important when you begin working with more complex sketches.

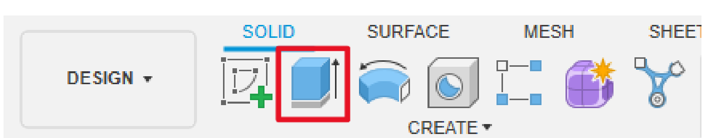
10. In the top right corner, click Finish Sketch.



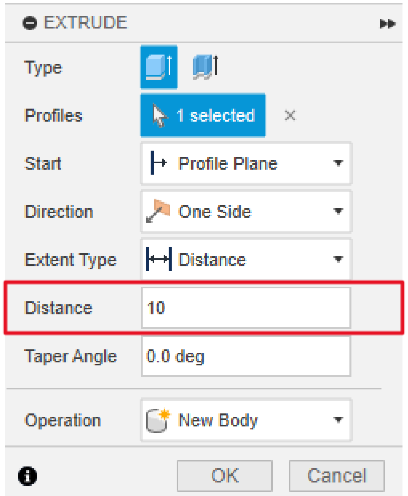
Making the Sketch 3D

Once you finish your sketch, you will want to turn it into a three-dimensional part – Fusion 360 provides several features to do so. You can add depth to your sketch with **Extrude**, revolve a sketch around a selected axis with **Revolve,** or even create a shape that connects two separate sketches with **Loft**. Depending on what you are tasked with designing, you might use one feature over all others. This tutorial will focus on the extrusion feature as it is the most widely used.

1. Click Extrude under the Solid tab or press E. Your complete sketch will turn blue.



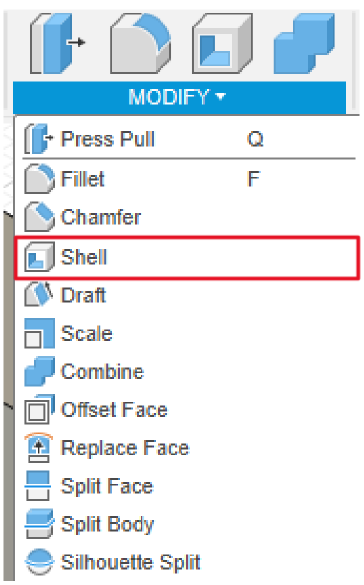
2. Click the Distance box and type '10'. You will see a thin three-dimensional rectangle.



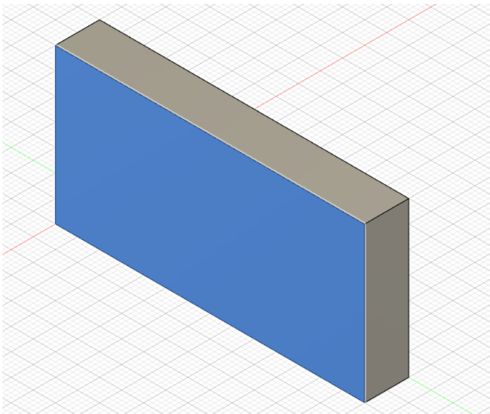
Modifying the Solid Body

Once your sketch is turned into a three-dimensional object, we typically refer to it as a **solid body**. The solid body can be modified in many useful ways. You can create beveled edges using **Chamfer,** rounded edges use **Fillet**, hollow out the body with **Shell,** or even scale the body by a scale factor using **Scale.** This tutorial will showcase the shell and fillet tools specifically.

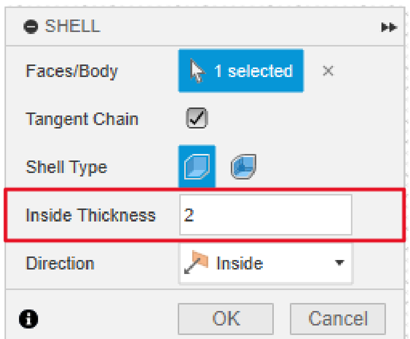
1. At the top of the screen, click Modify → Shell.



2. Click one of the front faces of the solid body. The face will turn blue.

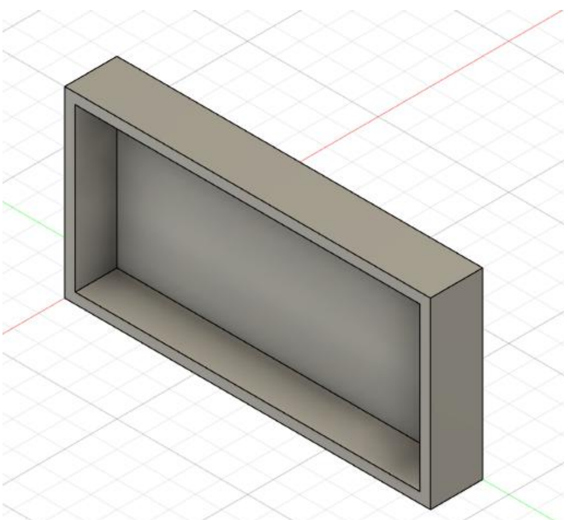


3. In the shell menu, click Inside Thickness and type ‘2’.

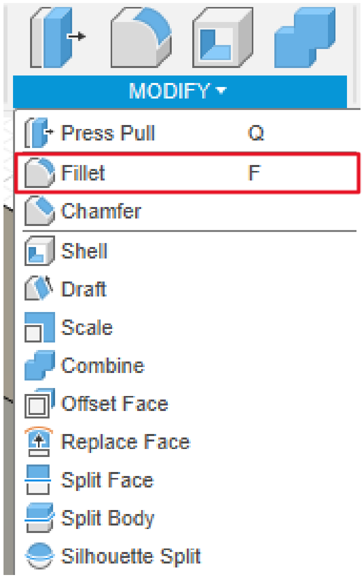


- *Note: The value of the inside thickness must be less than the extrusion depth we chose when creating the solid body.*

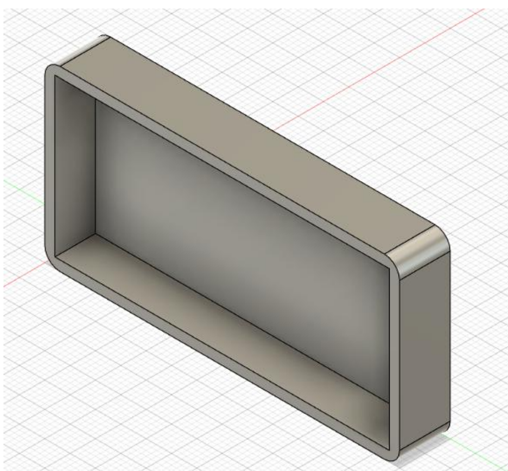
4. Click OK or hit enter. The front of your solid body will be hollow.



5. Click on Modify → Fillet. You can also hit ‘F’ to select the fillet tool.

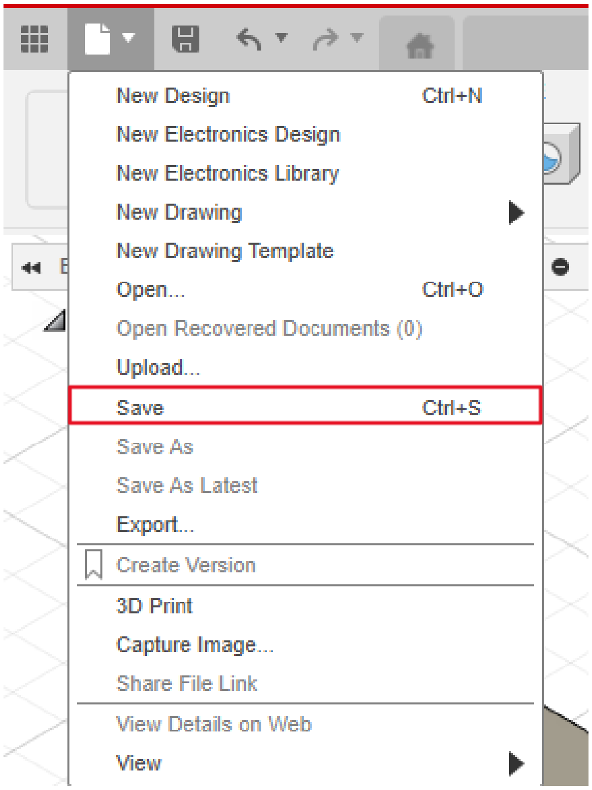


6. With the fillet tool selected, click the four outer vertical edges along the corners of the solid body. Once you have selected all four edges, type ‘3’ and hit Enter. The four edges of your part will be rounded.

7. 

- *Tip: To rotate the part, you can use the view cube in the top right corner. If you are using a mouse, you can rotate it by sliding your mouse while simultaneously holding Shift and the button on your scroll wheel (the middle mouse button).*

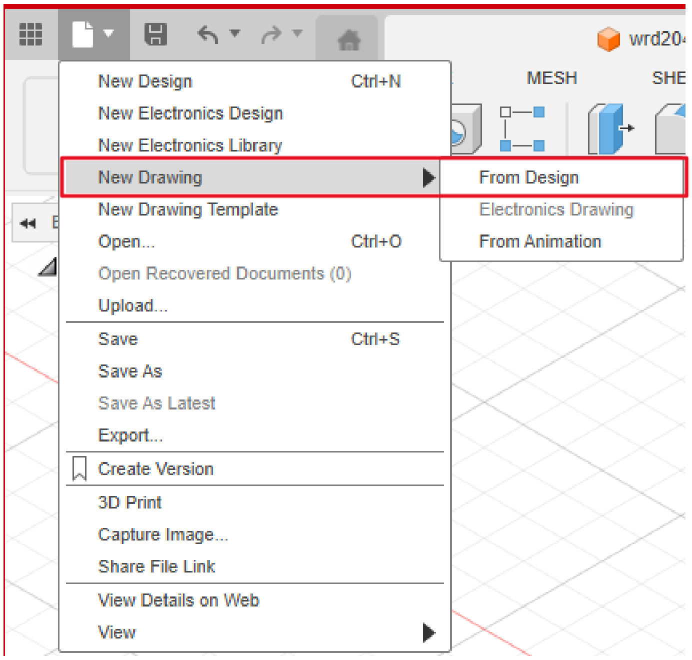
8. In the top left corner, click the paper icon → Save. Name your part and hit Enter.



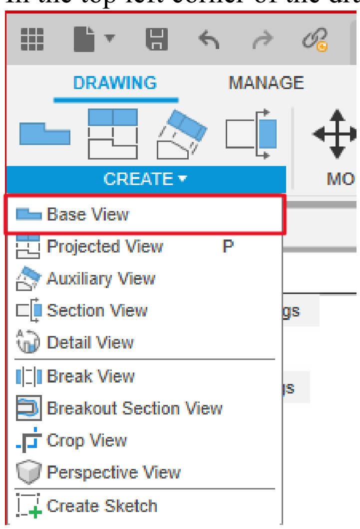
Documenting the Part

The process of documenting your finished part is the final and arguably the most important step. The documentation acts as a set of instructions that a manufacturer, fellow engineer, or client must read and interpret in order to recreate your design -- the drawing is *fundamentally equivalent* to the instructions you have viewed to sketch and model the part. Without proper documentation of the part, it would be impossible to know all the important details of a part just be looking at it in three dimensions. The documentation will include information like exact dimensions (sketch dimensions, fillet size, shell thickness), notes on materials used, or different views of the part. Documenting a part can be a meticulous process, especially as your parts become more complex, but it is of paramount importance in the engineering process.

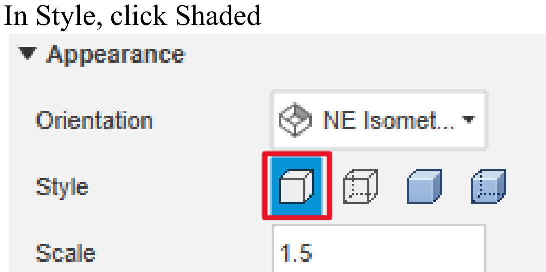
1. Click the paper icon → New Drawing → From Design. Click the solid body and click ‘OK’.



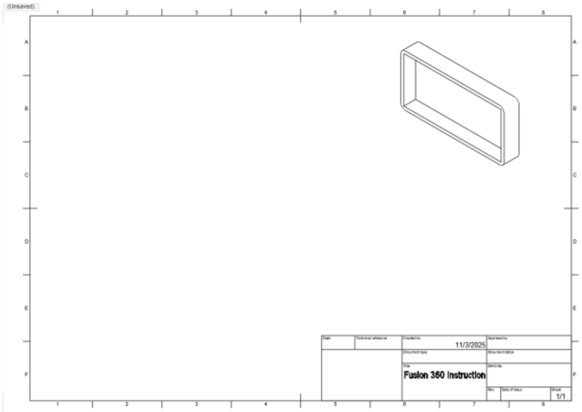
2. In the top left corner of the drawing file, click Create → Base View



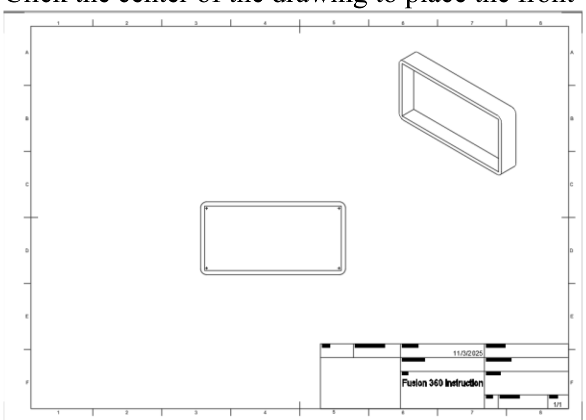
3. In the Drawing View menu under Appearance, click Orientation → NE Isometric
4. In Style, click Shaded



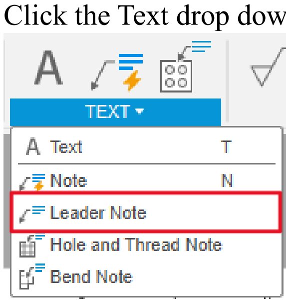
5. Click the Scale box and type ‘1.5’.
6. Click the right corner of the drawing to place the isometric view. Click ‘OK’.



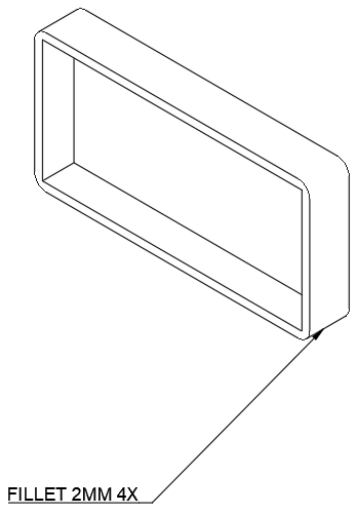
7. Click Create → Base View
8. Click Orientation → Front. Edit the style and scale to the same selections as in the NE Isometric view.
9. Click the center of the drawing to place the front view. Click ‘OK’.



10. Click the Text drop down → Leader **Note.**

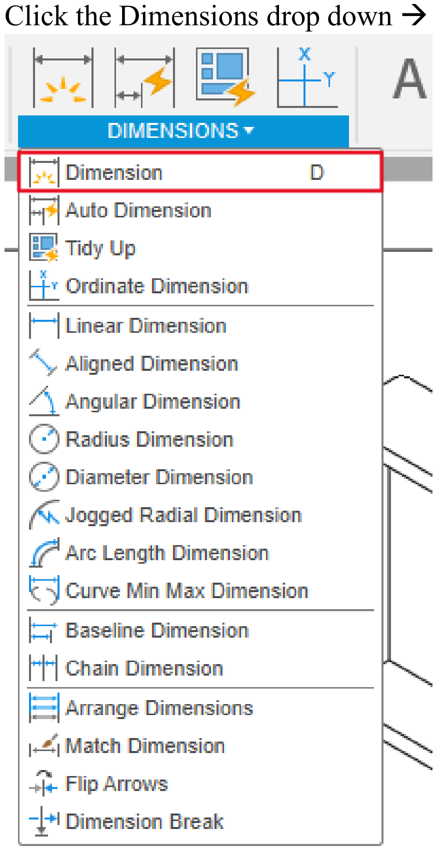


11. Click the bottom right corner of the NE Isometric view. Drag the arrow below, click, and type “FILLET R2 (4X). Hit Enter.

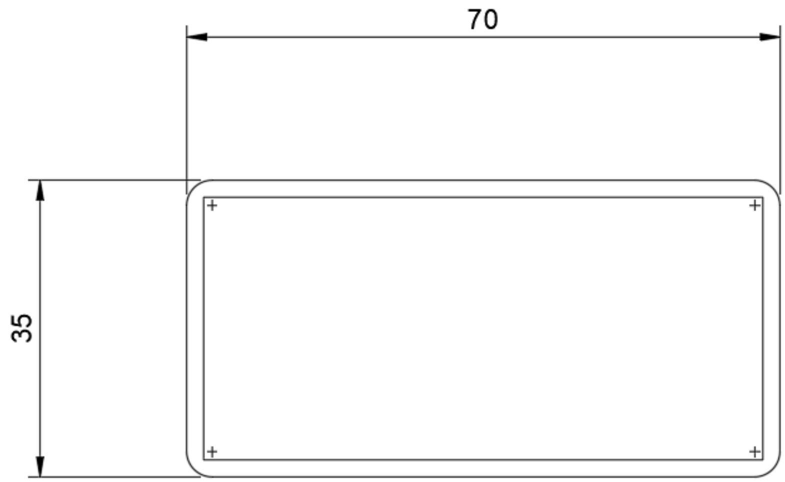


➢ *Note: This specifies to those reading your documentation that this edge was filleted with a radius of 2 mm. Adding "4X" tells us it was repeated to the other three alike edges as well – this reduces clutter and repetitiveness.*

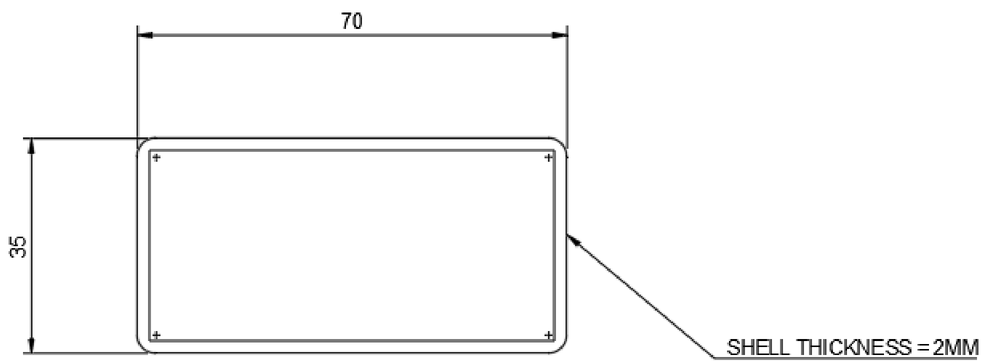
12. Click the Dimensions drop down → Dimension or press 'D'.



13. On the front view, click the left-most and right-most edge. Drag the dimension up and click. You will see 70 above the dimension.
14. On the same view, click the top and bottom edge. Drag the dimension to the left and click. You will see 35 next to the dimension.



15. Click Text dropdown → Leader Note. Click the right-outer edge of the front view and type "SHELL THICKNESS = 2MM".



16. Save the drawing